

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

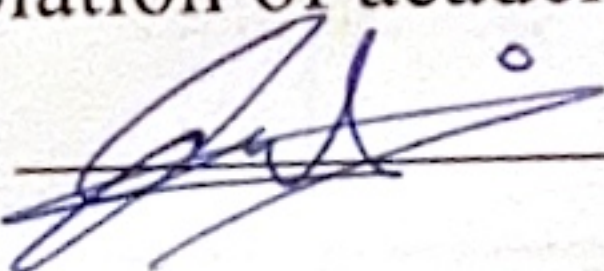
COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct (192511302)
I R. Ram Nishesh (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: 

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

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Course : Cloud Computing and Big data Analytics

Course code : CSA1512

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CO2 - AT-3 Application based Question

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand effectively. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answers to real-world e-commerce scenarios.



Introduction

* During seasonal sales such as Black Friday or festive shopping events, e-commerce websites experience sudden spikes in user traffic. Cloud computing provides auto scaling and load balancing to maintain application performance and availability.

Auto Scaling

* It automatically increases or decreases computing resources based on traffic demand.

Working

* During high traffic, additional virtual machines or containers are launched.

* During low traffic, unnecessary resources are removed.

x. A company x

* Resources are allocated automatically
manual intervention

Benefits

- * Prevents server overload
- * Maintains application performance
- * optimizes cloud resource usage
- * Reduce optional costs

Load balancing

* A load balancer distributes incoming requests evenly across multiple servers.

Functions

- * Prevents any single server from becoming overloaded
- * Redirects traffic if a server fails.
- * Ensures continuous service availability

Real world example

* During the Amazon Great Indian Festival Sale, millions of customers access the website simultaneously. Autoscaling launches additional servers, while Load balancers distribute customer requests across multiple instances. Monitoring tools continuously monitor system health & automatically adjust resources, ensuring uninterrupted shopping.

Conclusion

* Cloud Auto Scaling, load balancing & monitoring work together to provide high availability, minimize downtime & deliver an excellent user experience during traffic spikes.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantage compared to traditional storage systems. Provide an example of a disaster recovery use case.
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Introduction

* cloud providers achieve data durability by storing multiple copies of backup data across different storage devices and geographical locations. This data redundancy ensures that even if one storage device or data center fails, another copy of the data remains available.

Explanation

* To protect sensitive information, cloud providers use encryption both at rest and during data transmission. Identity and Access Management (IAM) role-based Access Control (RBAC) and multifactor authentication ensures that only authorized users can only access backup.

* Compared to traditional storage system, cloud storage eliminates the need to purchase expensive hardware and maintain physical infrastructure. It follows a pay as you go pricing model, allowing organizations to pay only for the storage they use while easily expanding storage capacity when required.

Example

* If a company's data center is damaged due to flood, fire or cyberattack, the backup stored securely in the cloud can be restored quickly. This enables the organization to recover its application and continue business operation with minimal downtime.

Conclusion

* cloud storage offers secure, reliable & cost effective backup solutions by combining durability, redundancy, making it an ideal choice for modern organizations.

3 Content delivery network (CDN)

Introduction

* A CDN is a distributed network of servers that delivers website and application content from location closest to user. It improves application performance by reducing latency and ensuring faster content delivery across globe.

Explanation

* CDNs store cached copies of web content on edge servers located in different regions. When users request data, the CDN delivers content the nearest edge server instead of the central server.

* cloud providers support global availability through multiple data centers, automatic failover and load balancing, ensuring continuous service even if one region experiences failure.

* This results in faster loading, lower latency & a better user experience.

Example

* Netflix uses CDNs to stream videos from nearby edge servers, reducing buffering.
Similarly, online multiplayer games use CDNs to reduce lag and provide smooth gameplay.

Conclusion

* CDNs improve application performance, reduce latency and provide reliable global access, making them essential for streaming, gaming and other real-time applications.

4 Cloud Security and Compliance

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Introduction

* Financial institutions require strong security and compliance to protect customer information and meet legal regulations. Cloud providers offer several security features to ensure safe cloud usage.

Explanation

* Cloud Providers comply with standards such as ISO 27001, PCI DSS and other regulatory requirements.

* Logging records all user and system activities while monitoring detects suspicious behaviour & security threats in real time.

Example

* A bank encrypts online payment transactions, stores customer information in an Indian cloud.

region and maintains ^{communication} to monitor transaction and detect fraudulent activities

conclusion

* Cloud security features such as logging, monitoring, encryption and regional data storage help financial institution securely store sensitive information and meet regulatory requirements

5 Containers and Kubernetes

Introduction

* Containerization is a modern application architecture with all required dependencies. It offers better performance and resource utilization than traditional VMs

Explanation

* Containers share the host OS, making them lighter and faster to start than VMs which require separate OS

* They consume less CPU, memory & storage while providing consistent performance

* Kubernetes automates container deployment, scaling, load balancing & failure recovery, making application management easier

Example

* An online shopping application can be divided into microservices such as user service, product service, order service. Each service runs in its own container while Kubernetes manages deployment, scaling & communication between them

Conclusion

* Containers and Kubernetes provide faster deployment, efficient resource utilization, easy scalability and portability, making them ideal for developing and managing modern cloud native applications